

## Cel:

192.168.200.0/24 - podzielić na połączenia między routerami

hasło dostępowe konsola,telnet: letmein

hasło na enable: t@jne

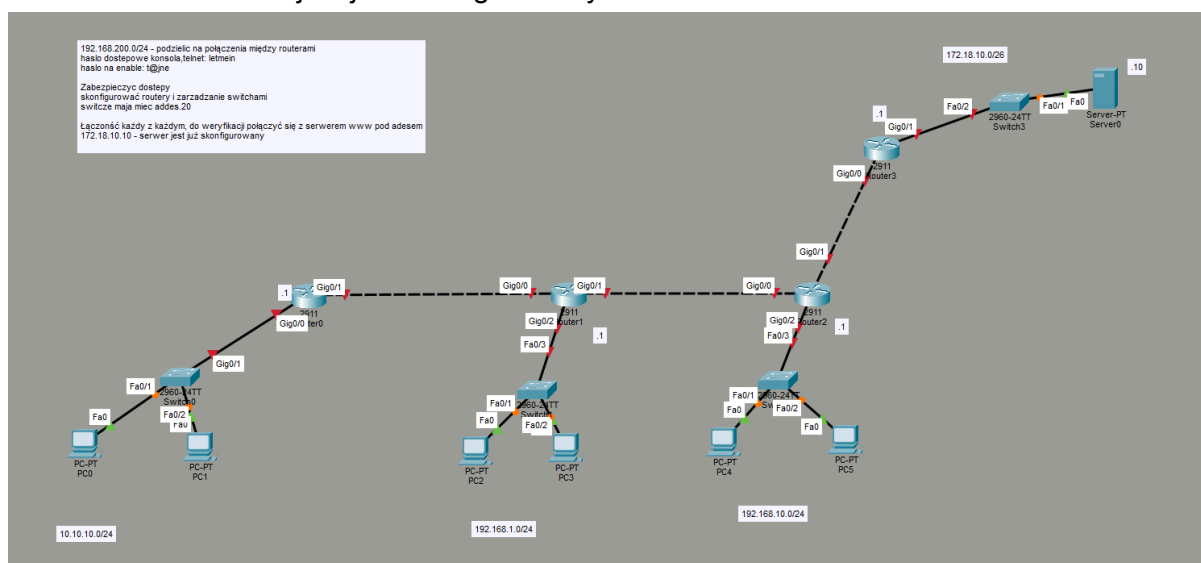
Zabezpieczyć dostępy

skonfigurować routery i zarządzanie switchami

switchy mają mieć adres.20

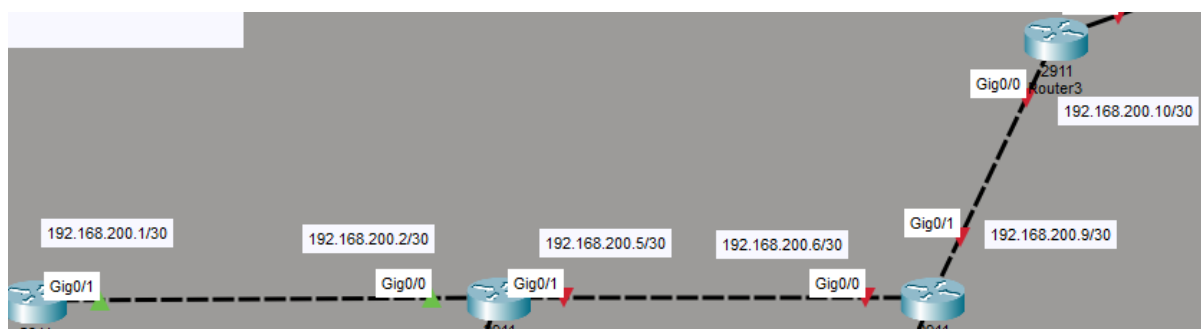
Łączyć każdy z każdym, do weryfikacji połączyć się z serwerem www pod adresem

172.18.10.10 - serwer jest już skonfigurowany



## Rozwiązanie:

podzielenie podsieci dla routerów:



Router0:

```

Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R0
R0(config)#line 0
No physical hardware support for line 3
R0(config)#line console 0
R0(config-line)#password letmein
R0(config-line)#login
R0(config-line)#exit
R0(config)#enable secret t@jne
R0(config)#int g0/0
R0(config-if)#ip add 10.10.10.1 255.255.255.0
R0(config-if)#no shut

R0(config-if)#int g0/1
R0(config-if)#ip add 192.168.200.0 255.255.255.254
Bad mask /31 for address 192.168.200.0
R0(config-if)#int g0/1
R0(config-if)#ip add 192.168.200.1 255.255.255.252
R0(config-if)#no shut

R0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
router ospf 1
R0(config-router)#network 10.10.10.0 0.0.0.255 area 0
R0(config-router)#pass
R0(config-router)#passive-interface gig0/0
R0(config-router)#network 192.168.200.0 0.0.0.3 area 0
R0(config-router)#

```

switch0:

```

Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW0
SW0(config)#line console 0
SW0(config-line)#password letmein
SW0(config-line)#login
SW0(config-line)#exit
SW0(config)#enable secret t@jne

SW0(config)#interface vlan 1
SW0(config-if)#ip add 10.10.10.20 255.255.255.0
SW0(config-if)#no shut

SW0(config-if)#exit
SW0(config)#ip defa
SW0(config)#ip default-gateway 10.10.10.1

```

router1:

```

Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#line console 0
Router(config-line)#password letmein
Router(config-line)#login
Router(config-line)#exit
Router(config)#enable secret t@jne
Router(config)#hostname R1
R1(config)#int g0/0
R1(config-if)#ip add 192.168.200.2 255.255.255.252
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#int g0/2
R1(config-if)#ip add 192.168.1.1 255.255.255.0
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

R1(config-if)#int g0/1
R1(config-if)#ip add 192.168.200.6 255.255.255.252
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

R1(config-if)#router ospf 1
R1(config-router)#pass
R1(config-router)#passive-interface g0/2
R1(config-router)#network 192.168.200.0 0.0.0.3
% Incomplete command.
R1(config-router)#network 192.168.200.0 0.0.0.3 area 0
R1(config-router)#network 192.168.200.5 0.0.0.3 area 0
00:22:01: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.200.1 on GigabitEthernet0/0 from LOADING to
FULL, Loading Done

R1(config-router)#network 192.168.1.0 255.255.255.0
% Incomplete command.
R1(config-router)#network 192.168.1.0 255.255.255.0 area 0
R1(config-router)#

R1(config-router)#int g0/1
R1(config-if)#no ip add 192.168.200.6 255.255.255.252
R1(config-if)#ip add 192.168.200.5 255.255.255.252
R1(config-if)#no shut

```

^poprawka^

switch1:

```

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#line console 0
Switch(config-line)#password letmein
^
% Invalid input detected at '^' marker.

Switch(config-line)#password letmein
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#enable secret t@jne
Switch(config)#int vlan1
Switch(config-if)#ip add 192.168.1.20 255.255.255.0
Switch(config-if)#no shut

Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Switch(config-if)#exit
Switch(config)#hostname SW1
SW1(config)#ip defa
SW1(config)#ip default-gateway 192.168.1.1
SW1(config)#

```

router2:

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#line console 0
Router(config-line)#password letmein
Router(config-line)#login
Router(config-line)#exit
Router(config)#enable secret t@jne
Router(config)#hostname R2
R2(config)#int g0/0
R2(config-if)#ip add 192.168.200.7 255.255.255.252
Bad mask /30 for address 192.168.200.7
R2(config-if)#ip add 192.168.200.7 255.255.255.256
^
% Invalid input detected at '^' marker.

R2(config-if)#ip add 192.168.200.6 255.255.255.252
R2(config-if)#int g0/2
R2(config-if)#ip add 192.168.10.1 255.255.255.0
R2(config-if)#int g0/1
R2(config-if)#ip add 192.168.200.9 255.255.255.252
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

```

```

R2(config-if)#int g0/0
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2(config-if)#int g0/2
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

R2(config-if)#router ospf 1
R2(config-router)#passive
R2(config-router)#passive-interface g0/2
R2(config-router)#network 192.168.10.0 0.0.0.255
% Incomplete command.
R2(config-router)#network 192.168.10.0 0.0.0.255 area 0
R2(config-router)#network 192.168.200.4 0.0.0.3 area 0
R2(config-router)#network 192.168.200.8 0.0.0.3 area 0
00:33:10: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.200.6 on GigabitEthernet0/0 from LOADING to
FULL, Loading Done

R2(config-router)#

```

## Switch2:

```

Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#line console 0
Switch(config-line)#password letmein
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#enable secret t@jne
Switch(config)#int vlan 1
Switch(config-if)#ip add 192.168.10.20 255.255.255.0
Switch(config-if)#no shut

Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Switch(config-if)#exit
Switch(config)#hostname SW2
SW2(config)#ip def
SW2(config)#ip default-gateway 192.168.10.1
SW2(config)#

```

## Router3:

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#line console 0
Router(config-line)#password letmein
Router(config-line)#login
Router(config-line)#exit
Router(config)#enable secret t@jne
Router(config)#int g0/0
Router(config-if)#ip add 192.168.200.10 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#int g0/1
Router(config-if)#ip add 172.18.10.1 255.255.255.192
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#hostname R3
R3(config)#
R3(config)#router ospf 1
R3(config-router)#network 192.168.200.8 0.0.0.3 area 0
R3(config-router)#network 172.18.10.0 0.0.0.
00:37:56: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.200.9 on GigabitEthernet0/0 from LOADnetwork
172.18.10.0 0.0.0.63
% Incomplete command.
R3(config-router)#network 172.18.10.0 0.0.0.63 area 0
R3(config-router)#

```

### Switch3:

```

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#line console 0
Switch(config-line)#password letmein
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#enable secret t@jne
Switch(config)#int vlan1
Switch(config-if)#ip add 172.18.10.20 255.255.255.192
Switch(config-if)#no shut

Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Switch(config-if)#exit
Switch(config)#ip def
Switch(config)#ip default-gateway 172.18.10.1
Switch(config)#hostname SW3
SW3(config)#

```

### PCty:

IP Configuration

☒ DHCP ☒ Static

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

IP Configuration

☒ DHCP ☒ Static

IPv4 Address

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IP Configuration

☒ DHCP ☒ Static

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

IP Configuration

☒ DHCP ☒ Static

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

testy lacznosci:

z pc0:

```
C:\>ping 192.168.1.30

Pinging 192.168.1.30 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.30: bytes=32 time<1ms TTL=126
Reply from 192.168.1.30: bytes=32 time<1ms TTL=126
Reply from 192.168.1.30: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

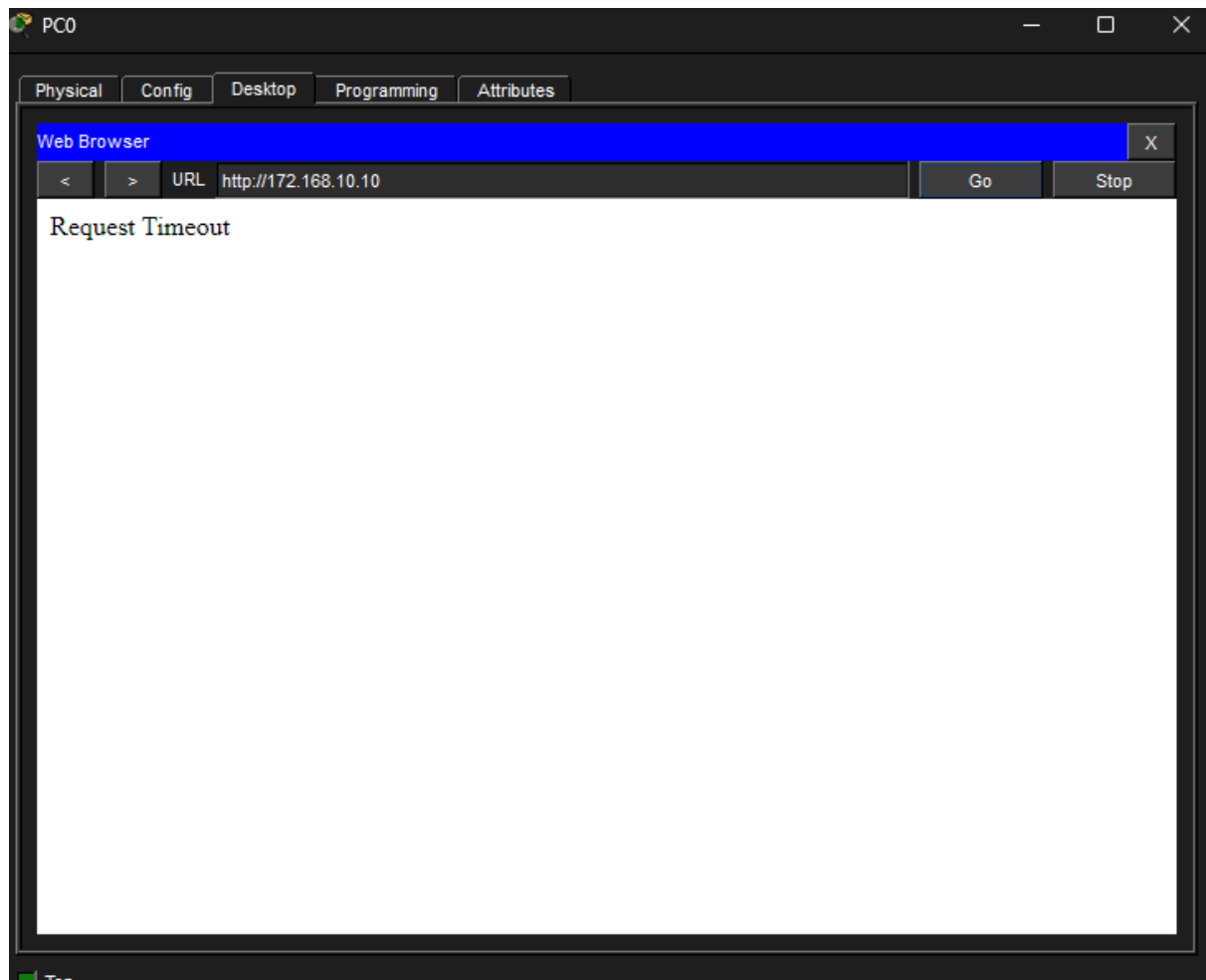
C:\>ping 192.168.10.30

Pinging 192.168.10.30 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.30: bytes=32 time<1ms TTL=125
Reply from 192.168.10.30: bytes=32 time<1ms TTL=125
Reply from 192.168.10.30: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.10.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

czyli pingi smigaja, sprawdzmy WWW



ok, nie działa, sprawdzmy pingi.

```

C:\>ping 172.18.10.10

Pinging 172.18.10.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.18.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.200.10

Pinging 192.168.200.10 with 32 bytes of data:

Reply from 192.168.200.10: bytes=32 time<1ms TTL=252
Reply from 192.168.200.10: bytes=32 time=10ms TTL=252
Reply from 192.168.200.10: bytes=32 time<1ms TTL=252
Reply from 192.168.200.10: bytes=32 time<1ms TTL=252

Ping statistics for 192.168.200.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
C:\>ping 172.18.10.1

Pinging 172.18.10.1 with 32 bytes of data:

Reply from 172.18.10.1: bytes=32 time<1ms TTL=252
Reply from 172.18.10.1: bytes=32 time=12ms TTL=252
Reply from 172.18.10.1: bytes=32 time=12ms TTL=252
Reply from 172.18.10.1: bytes=32 time<1ms TTL=252

Ping statistics for 172.18.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 6ms

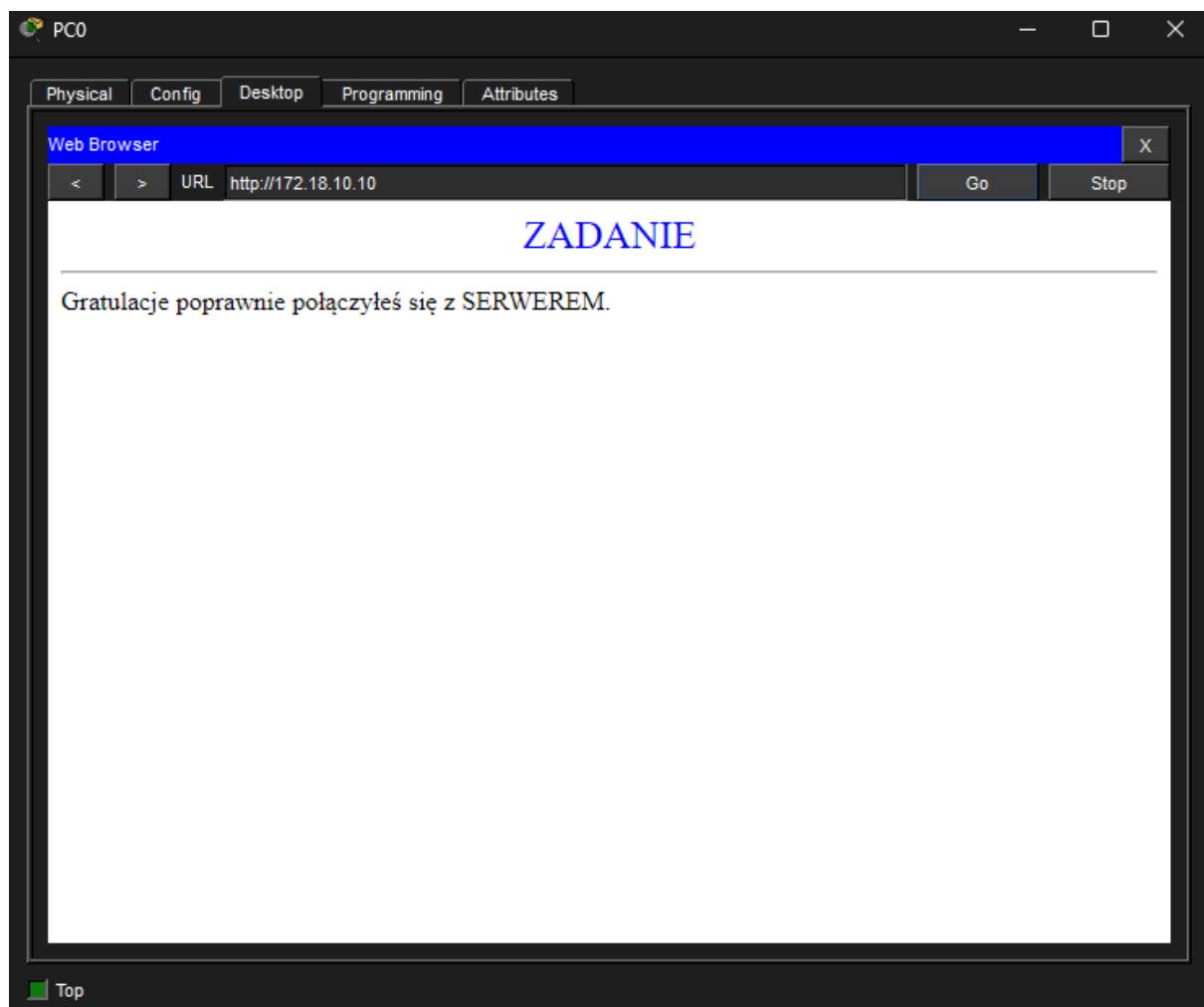
C:\>

```

do serwera nie dochodza, ale do routera bez problemu. sprawdzmy konfiguracje serwera ktorej nie mielismy dotykac.

| IP Configuration                      |                                         |
|---------------------------------------|-----------------------------------------|
| <input checked="" type="radio"/> DHCP | <input checked="" type="radio"/> Static |
| IPv4 Address                          | 172.18.10.10                            |
| Subnet Mask                           | 255.255.255.192                         |
| Default Gateway                       | 162.18.10.1                             |
| DNS Server                            | 0.0.0.0                                 |

i jest blad w zadaniu! zly gateway! zmieniamy i sprawdzamy czy dziala jak powinno:



działa, koniec zadania :D